

OFFICIAL SDLC SPECIFICATION

System Design & Architecture Document (SDD)

SmartFarm Management Application

Document Version: 1.0

SDLC Stage: Phase 2 — System
Architecture & Design

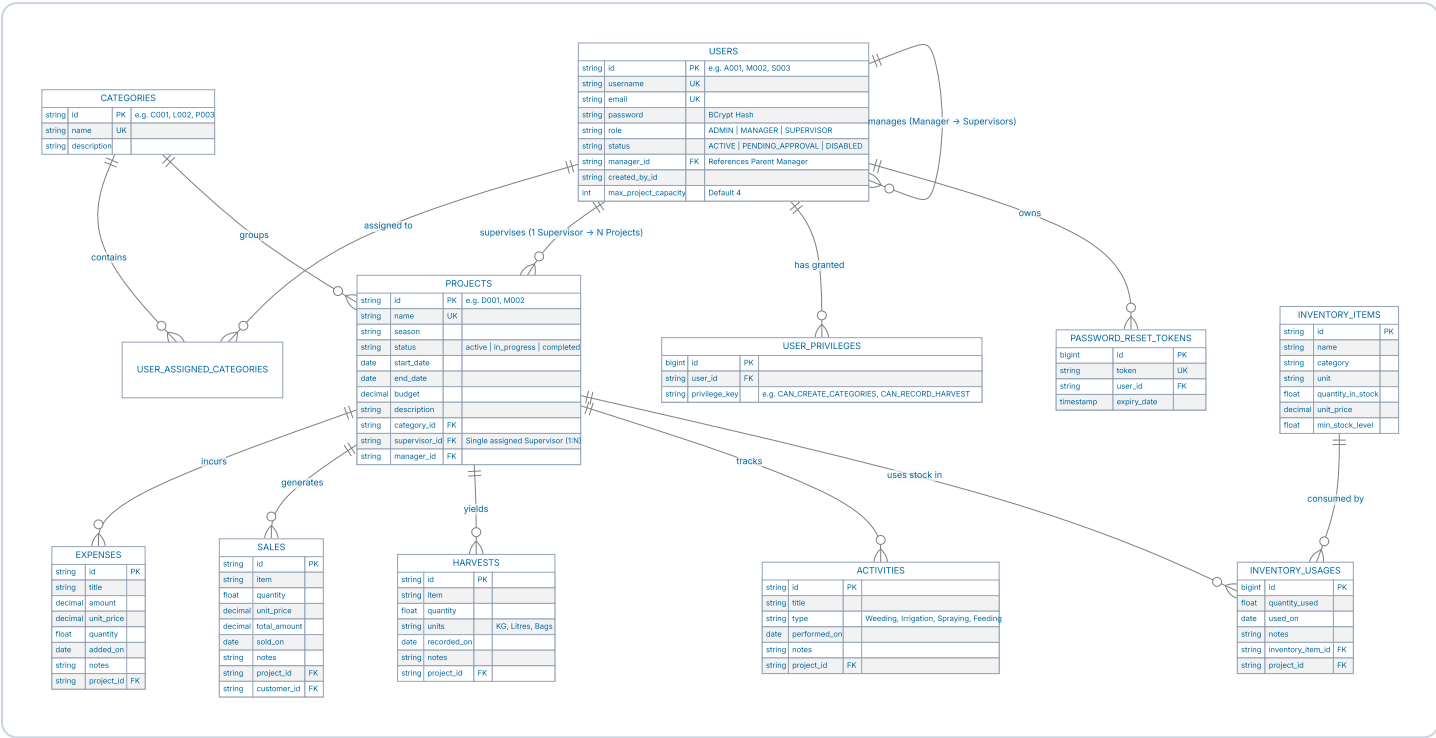
Status: Approved Technical Blueprint

1. Architectural Overview & Technology Stack

The SmartFarm system follows a **3-Tier Clean Architecture** separating presentation, business orchestration, and persistent storage:

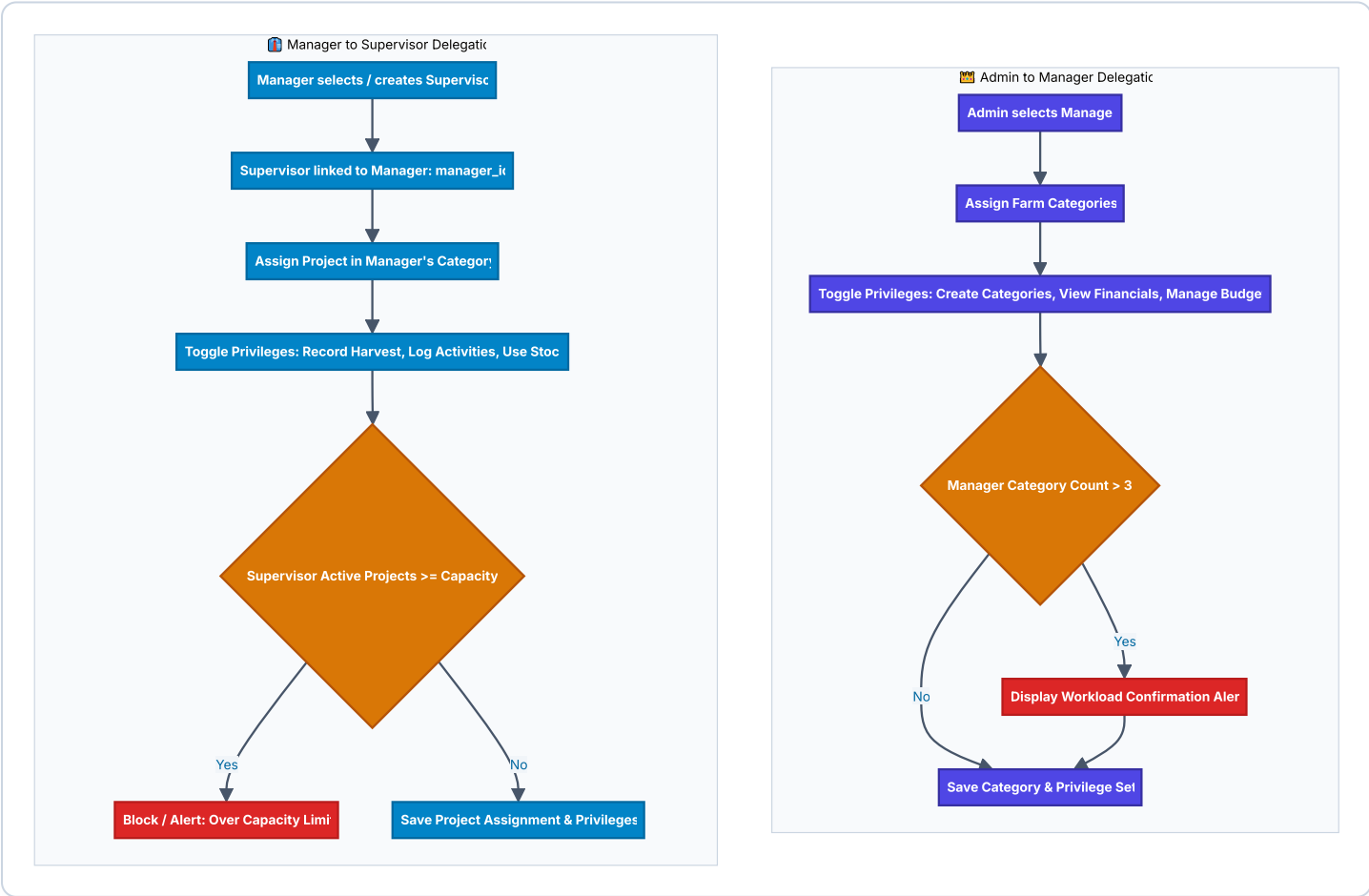


2. Complete Entity-Relationship Model (ERD)



3. Access Control & Privilege Delegation Engine (PBAC)

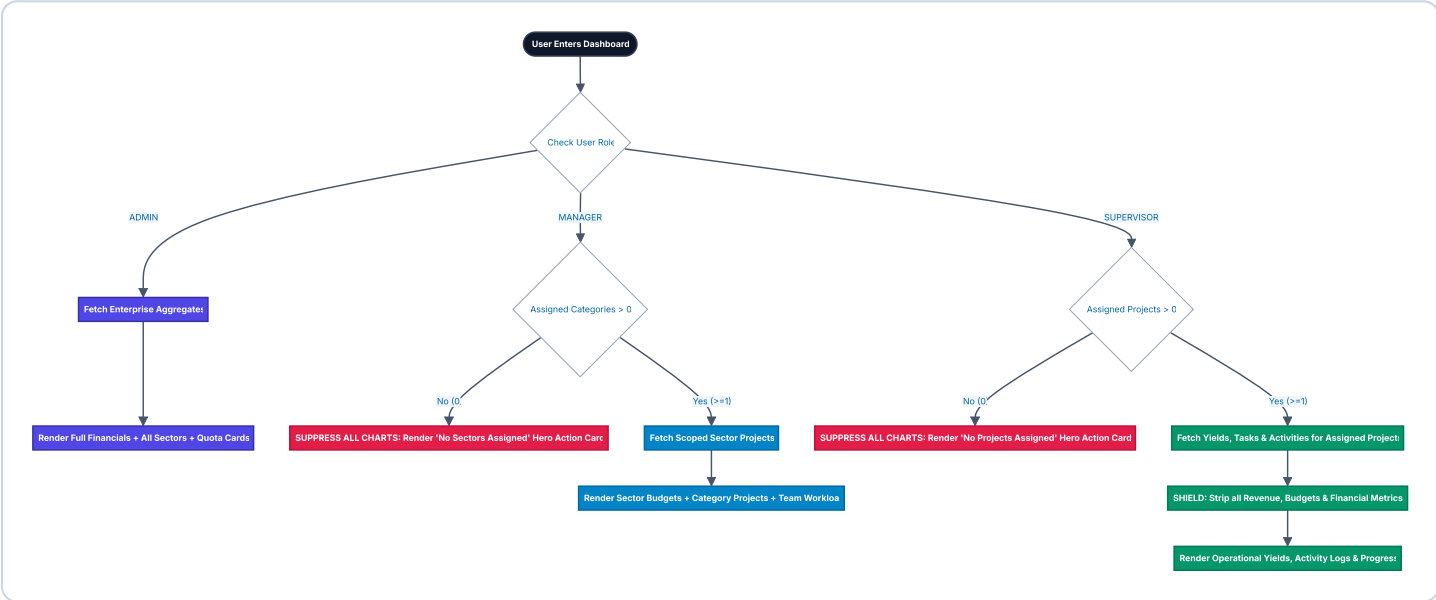
3.1 Delegation Hierarchy & Workload Guard



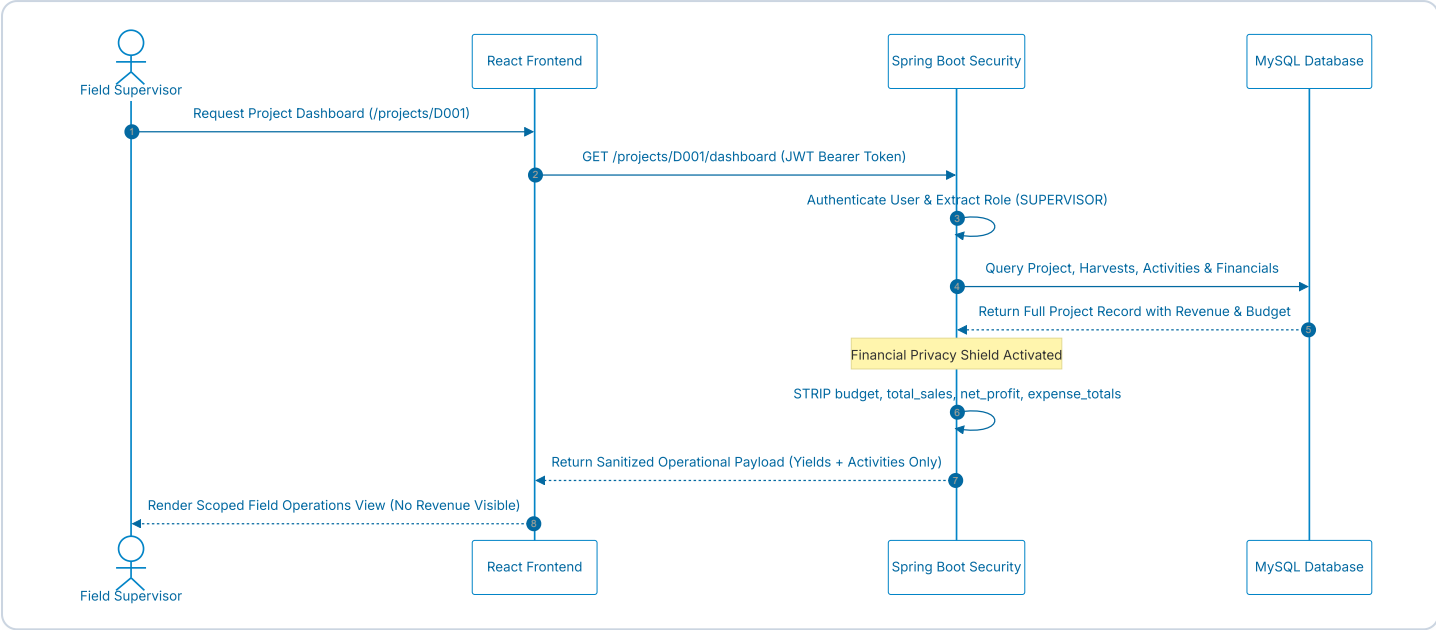
3.2 Privilege Definition Catalog

Privilege Key	Applied To	Backend Endpoint Guard	Default State
`CAN_CREATE_CATEGORIES`	`MANAGER`	`POST /categories`	Disabled (`false`)
`CAN_CREATE_SUPERVISORS`	`MANAGER`	`POST /users/staff` (Role = `SUPERVISOR`)	Enabled (`true`)
`CAN_VIEW_FINANCIALS`	`MANAGER`	`GET /projects/summary`, revenue fields	Enabled (`true`)
`CAN_MANAGE_BUDGETS`	`MANAGER`	`PUT /projects/{id}` (Budget property)	Disabled (`false`)
`CAN_RECORD_HARVEST`	`SUPERVISOR`	`POST /harvest`	Enabled (`true`)
`CAN_LOG_ACTIVITIES`	`SUPERVISOR`	`POST /activities`	Enabled (`true`)
`CAN_USE_INVENTORY`	`SUPERVISOR`	`POST /inventory/use`	Enabled (`true`)
`CAN_RECORD_EXPENSES`	`SUPERVISOR`	`POST /expenses`	Disabled (`false`)
`CAN_RECORD_SALES`	`SUPERVISOR`	`POST /sales`	Disabled (`false`)

4. Scoped Dashboard & Zero-State State Machine



5. Security & Financial Privacy Shield Sequence



6. Zero-Cost Cloud Deployment Topology

